

5CM510 – Network System development

Lecture 16: Software Defined Networks

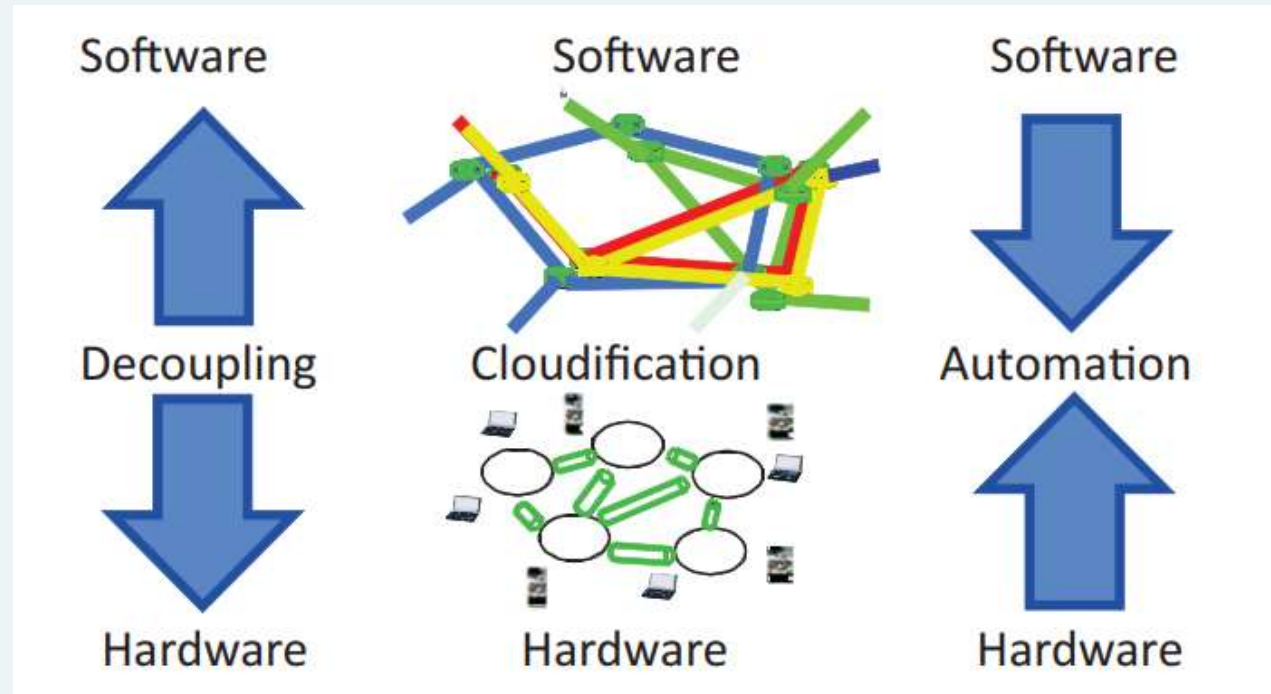
Lecture Outline

- The Objective of SDN?
- ONF, SDN Architecture, OPNFV
- Southbound and Northbound interface
- The Controller and Application layer
- Summary

Objective of SDN

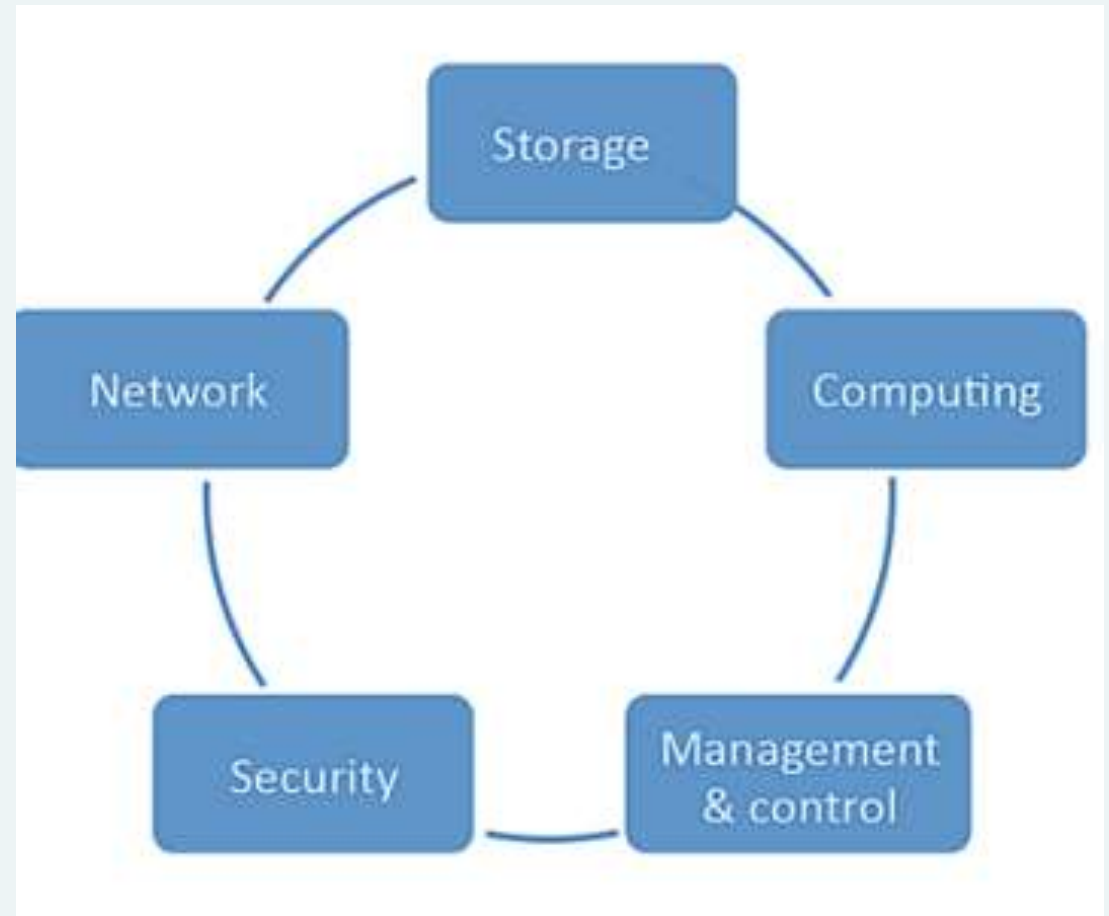
- The objective of SDN (Software-Defined Networking) is to reduce costs by virtualization, automation and simplification. For this purpose, SDN facilitates the customization of the networks, a very short set-up time and a network deployment with the right quality of service (QoS) rather than a general QoS.
- The architecture of SDN can be summarized with three fundamental principles.
- The first is the **decoupling of the physical and virtual layers (hardware and software)**. This enables virtual devices to be loaded on hardware machines provided, of course, that these hardware machines can host a hypervisor or containers.
- The second principle is going **from a hardware to a logical aspect**. This new environment enables us to spontaneously change the network by adding a new network or by taking a network away as much as we want.
- The third principle is that of **automation – the best possible automation – of the operations carried out on the network**, whether for **management or for control**.

Fundamental principles of SDN



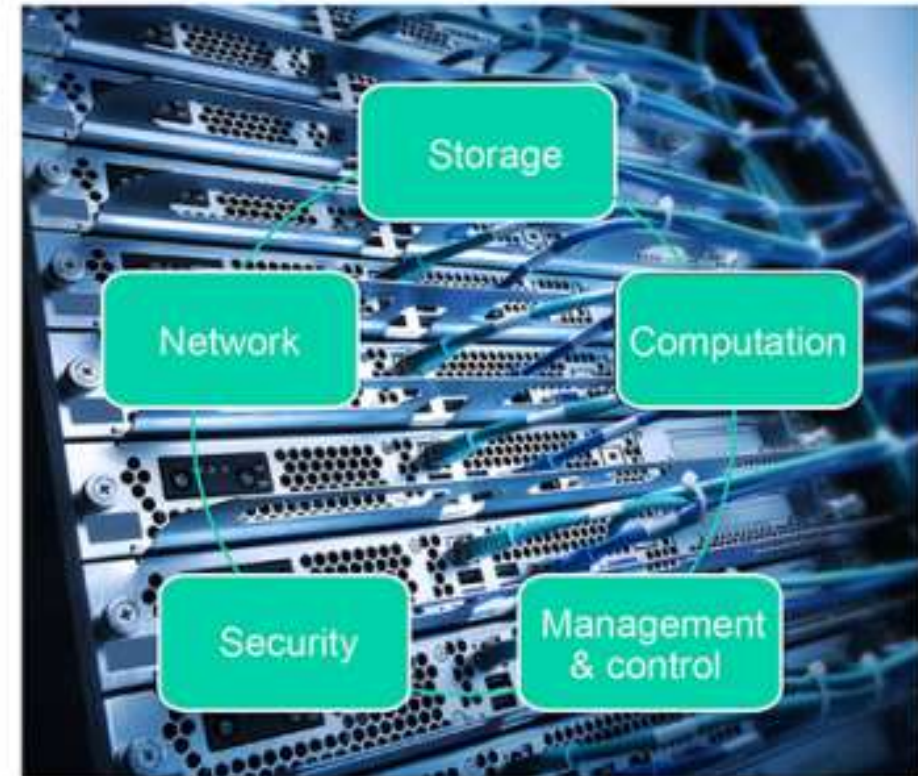
Five Domains

- These five domains are Network, Storage, Computing, Security, and Management/control.
- The storage, processing and the network use virtual machines associated with domains that share the hardware resources - on the same server.
- For the environment to be able to be executed without problems, we must add security, management and control virtual machines.
- Today, a company needs to have all five of these groups of virtual machines set up to constitute an operational information system.

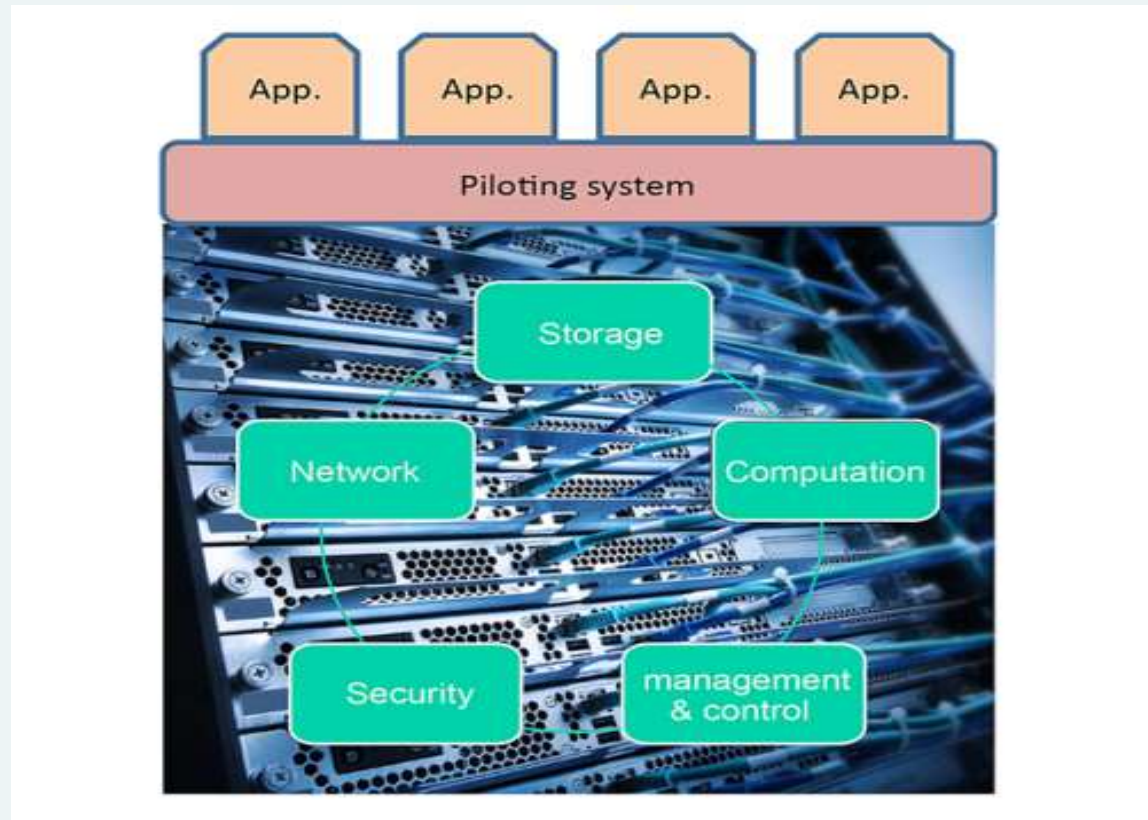


Virtualization of Five domains

- The five domains described earlier can be put in place by way of virtual machines associated with each of the domains. The whole computing environment of a company can thus be concentrated in the Cloud in the form of virtual machines distributed in data centres.



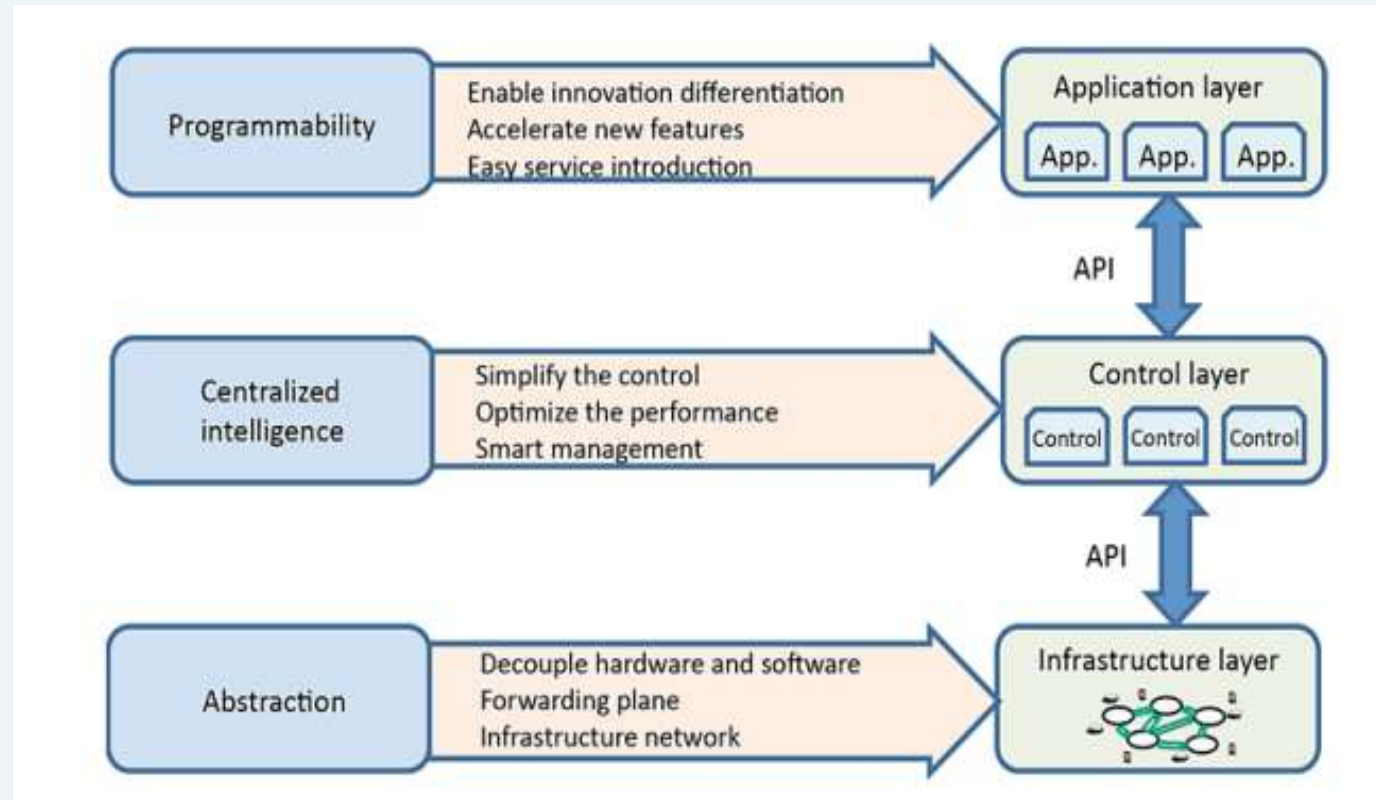
The Pilot program and Orchestration



ONF Architecture: SDN standardization

- This standardization was carried out by the ONF (Open Networking Foundation), which was set up under the auspices of large companies in the USA, following the proposal of this architecture by Stanford University and Nicira. The architecture proposed by the ONF comprises three layers.
- The **bottom layer** is an abstraction layer, which decouples the hardware from the software and is responsible for data transport. This level describes the protocols and algorithms that enable IP packets to advance through the network to their destination. This is called the infrastructure plane.
- The **middle layer** is the control plane. This plane contains the controllers providing control data to the data plane so that the data are channelled as effectively as possible. The ONF's vision is to centralize control to facilitate the recovery of a great deal of information on all the clients. The centralized controller enables obtaining a sort of intelligence.
- The **uppermost layer**, the application plane, is responsible for the applications needed by the clients and storage, computation, network, security and management applications. This layer introduces the programmability of the applications, and sends the controller all of the necessary information to allow the opening of the software networks meeting the needs of the applications

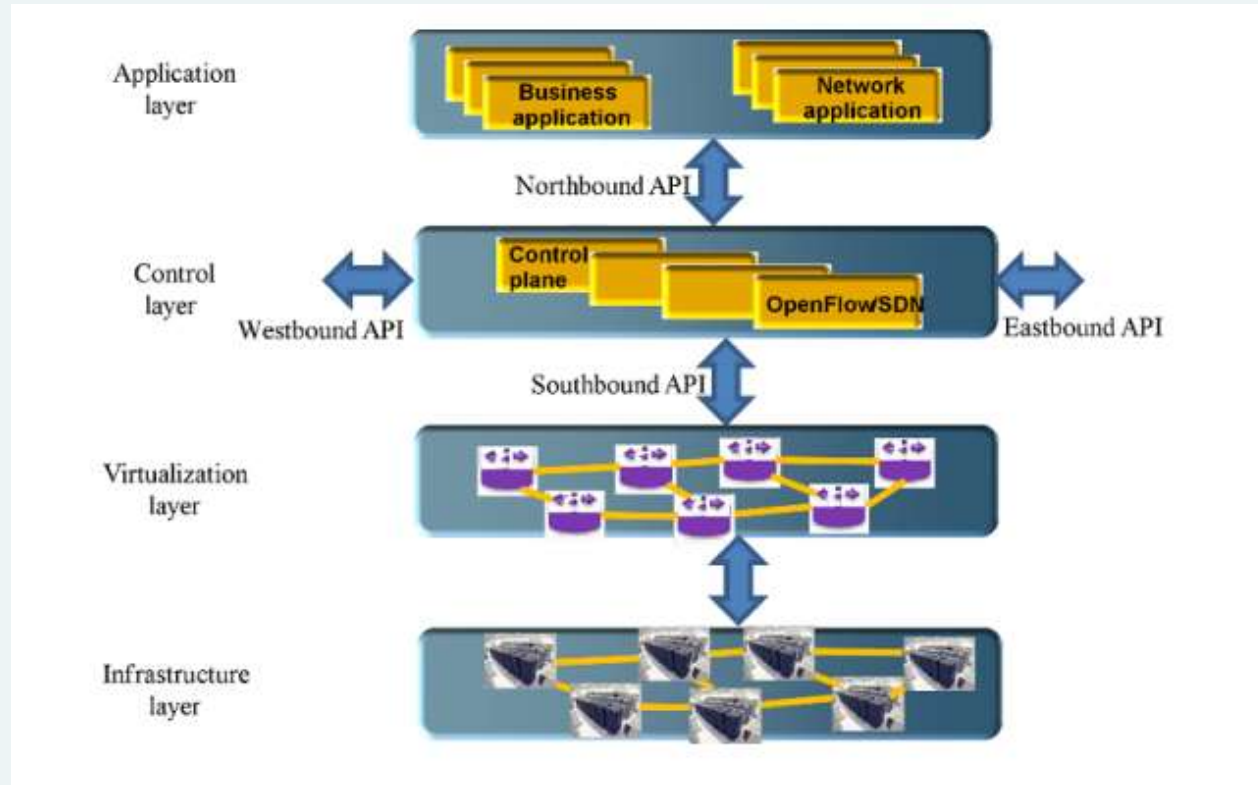
ONF Architecture



SDN's Physical and Logical planes

- The physical plane oversees all the hardware, and more generally, the physical infrastructure. The logical plane corresponds to the establishment of the software networks constructed based on virtual machines, sharing the physical infrastructure in accordance with the rules deriving from the higher layers.
- This vision of the architecture enables us to discern the hardware and the networks that exist in companies from the software, which is added to offer the necessary flexibility.
- This architecture requires data centres ranging in size from very small to very large, depending on the size of the company and on the resource's distribution to the periphery.
- Telecom operators have not missed this opportunity and have entered the market as Cloud providers.
- Companies such as Amazon and Google have gone directly for the goal, putting in place the infrastructure necessary to become major players in the world of telecommunications.

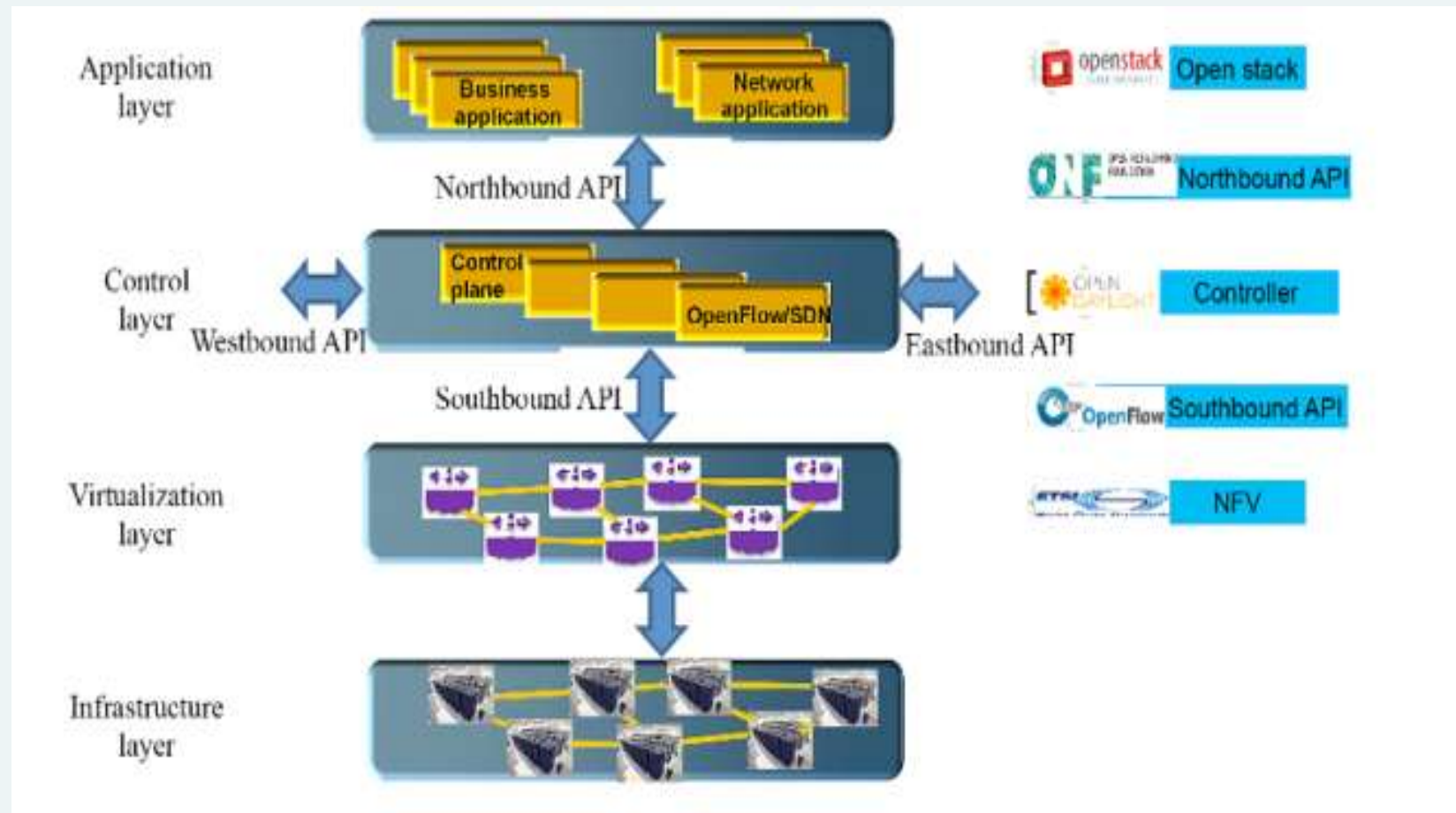
SDN Architecture



SDN Architecture

- The control layer and the application layer with the northbound and southbound APIs (Application Programming Interfaces) between those layers, and the eastbound and westbound APIs with other controllers.
- The **northbound interface** facilitates communication between the application level and the controller. Its purpose is to describe the needs of the application and to pass along the commands to orchestrate the network.
- The **southbound interface** describes the signalling necessary between the control plane and the virtualization layer. With this aim in mind, the controller must be able to determine the elements that will make up the software network to set up.
- The bandwidth necessary for the feeding back of these statistics may represent a few percent of the network's capacity, but this is crucial for optimization which will improve performance by much more than a few percent.
- The **eastbound interface** enables two controllers of the same type to communicate with one another and make decisions together. The **westbound interface** must also facilitate communication between two controllers, but ones which belong to different sub-networks.

Open-source developments



SDN: Open-source developments

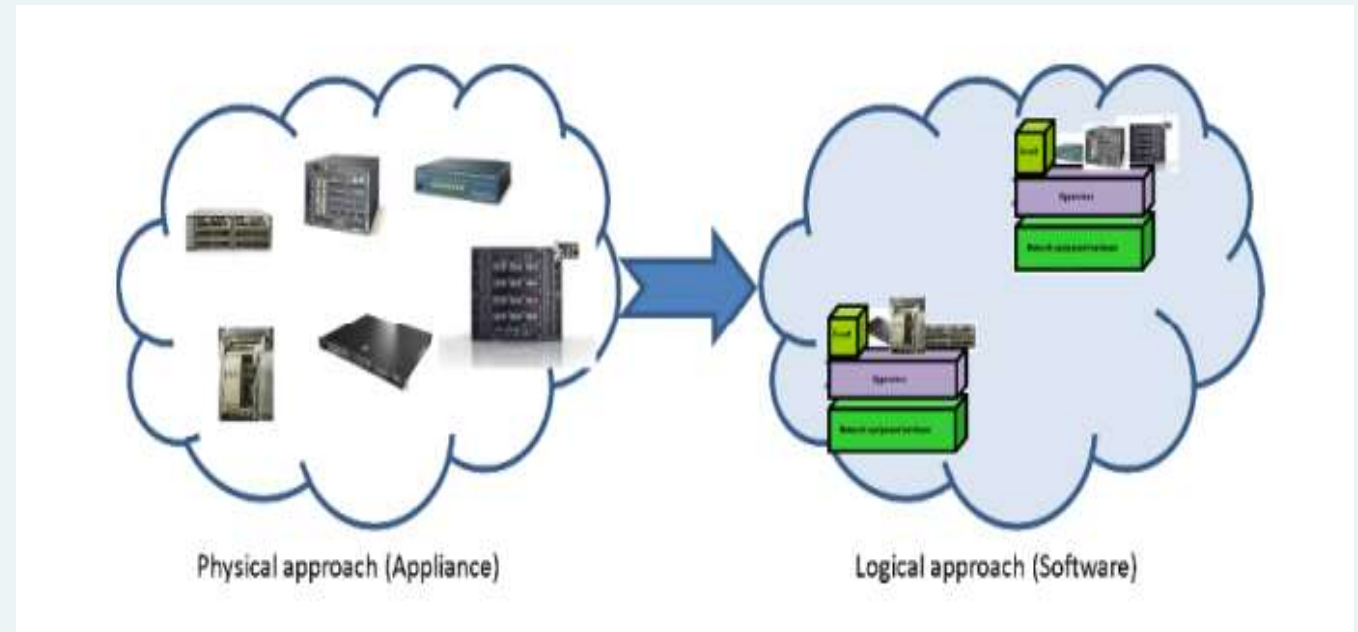
- Network virtual machines were standardized by the ETSI in a working group called NFV (Network Functions Virtualization) .
- NFV aims to standardize all network functions with a view to virtualizing them and facilitating their execution in different places from the original physical machine.
- To complete this standardization, an open-source software code has been developed which allows full compatibility between virtual machines.
- The control plane includes the controllers. One of the best known is OpenDaylight – an open-source controller developed collaboratively by numerous companies.
- This controller contains many modules, often developed in the interest of the particular company that carried out that work. Today, OpenDaylight is one of the major pieces in the Cisco architecture, as well as that of other manufacturers.
- There are many other controllers – open-source ones as well – such as OpenContrail, ONOS, Flood Light, etc.

SDN: Open-source developments

- The uppermost layer represents the Cloud management systems. It is roughly equivalent to the operating system on a computer. It includes OpenStack, which was the system which was most favoured by the developers, but many other products exist, both open source and proprietary.
- The southbound interface is often known through its standard naming from the ONF: OpenFlow. OpenFlow is a signalling system between the infrastructure and the controller. This protocol was designed by Nicira and has led to a de facto standard from the ONF.

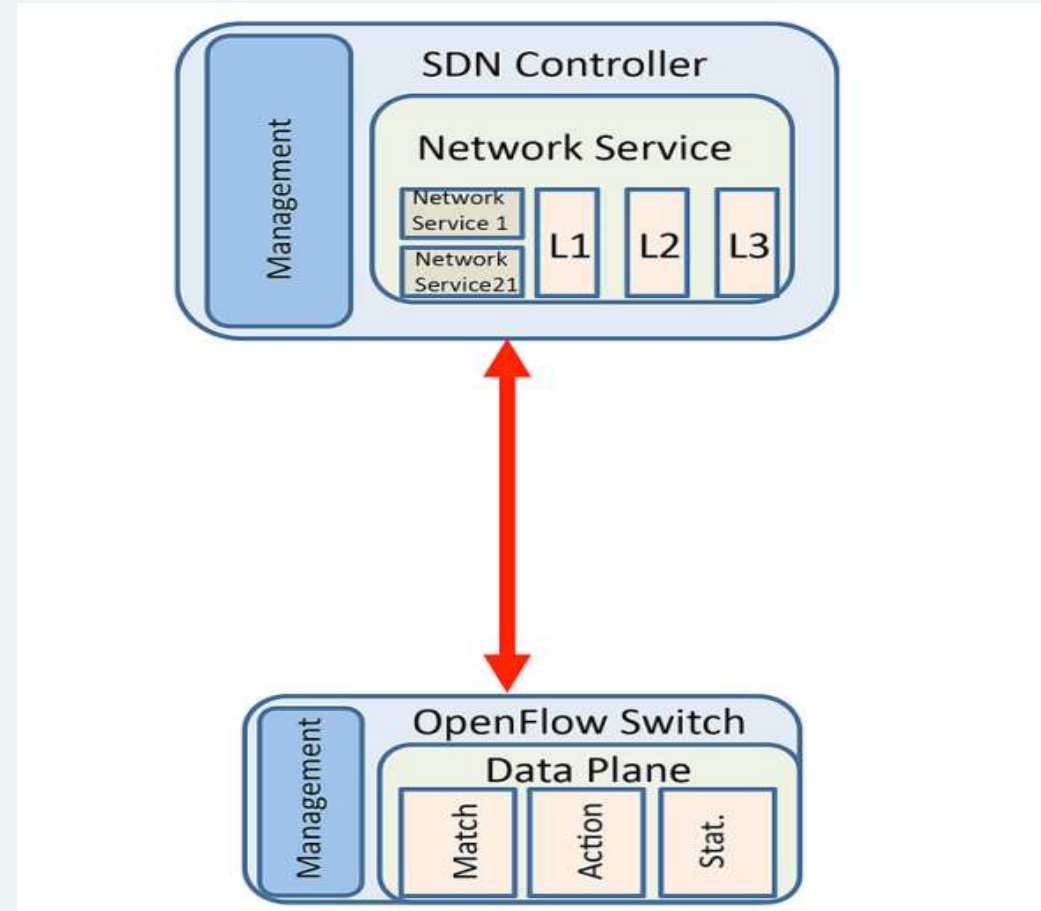
NFV

- The purpose of NFV (Network Functions Virtualization) is to decouple the network functions from the network equipment. This decoupling enables us to position the software performing the control of a device on a different machine than the device itself.
- OPNFV provides an NFV Infrastructure (NFVI), Virtualized Infrastructure Management (VIM) and APIs to other NFV elements. This collection forms the basic software necessary for Management and Network Orchestration (MANO).



Southbound Interface

- The southbound interface is situated between the controller and the devices on the virtualization plane.
- This signalling protocol (diagram) passes the configuration commands in one direction and manages the statistical information feedback in the other. There are many open-source proposals for some under development. An initial list of protocols for this southbound interface is:
- OpenFlow from the ONF;
- I2RS (Interface to Routing Systems) from the IETF;
- Open vSwitch Data Base (OvSDB);
- NetConf
- SNMP
- LISP
- BGP

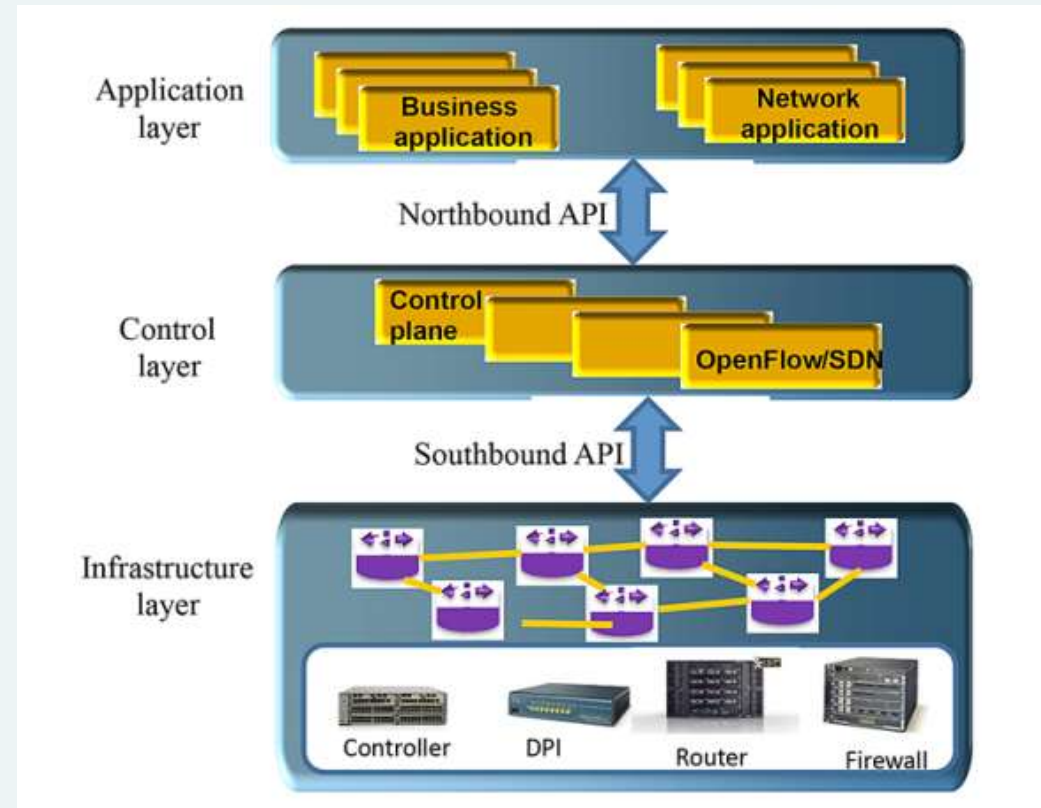


Southbound Interfaces: OpenFlow

- The data transported obey the trilogy “match-action-statistics” – in other words, we need to:
- Determine the streams uniquely by matching a certain number of elements such as addresses or port numbers;
- Specify the actions transmitted from the controller to the networking devices, such as updating a forwarding table or switch table, or, more generally, a forwarding device;
- Transfer tables containing statistics on the degree of use of the ports, transmission lines and nodes, such as the number of bytes sent through a given port.

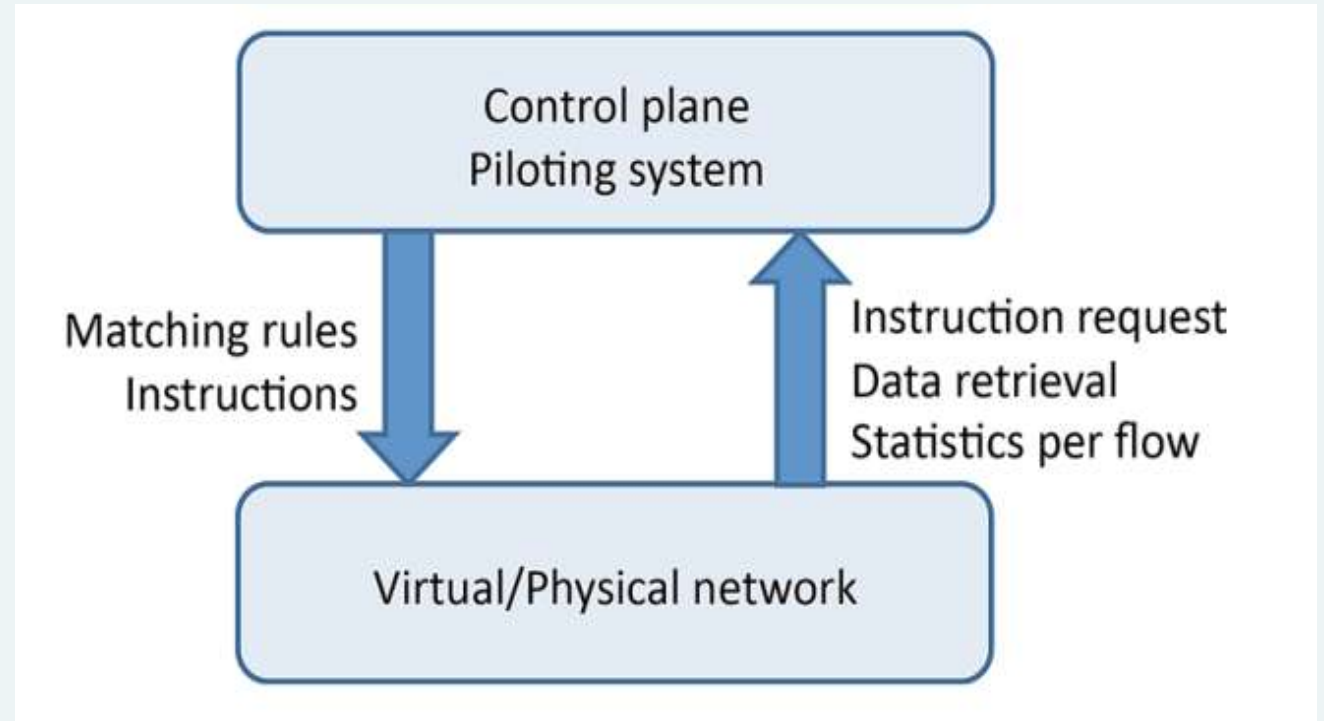
The Controller

- The controller, as its name indicates, is designed to control the data plane and receives from the application layer the necessary elements to determine the type of control that needs to be exerted.
- The controller must receive policy rules to be respected. Based on the description of the applications to be considered, it deduces the actions needed on the networking equipment.
- The actions can be carried out on routers, switches, firewalls, load balancers, virtual VPNs and on any other hardware that is necessary to the good functioning of the network.



The Controller function: Load balancing protocol

- Load balancing function decides the best paths to follow on the data plane.
- This module must decide, on the basis of the statistics received, which nodes in the network infrastructure the packets should be sent through.
- This decision should optimize the user demands or, more specifically, the user applications. Load balancing is essentially valid during peak hours.
- During other times, the load balancer must determine the most appropriate paths, on the basis of the users' requirements.
- Load balancing becomes load unbalancing: unlike at peak times, we need to channel as many streams as possible through common paths to be able to shut down the maximum possible number of intermediary nodes.



The Northbound Interface

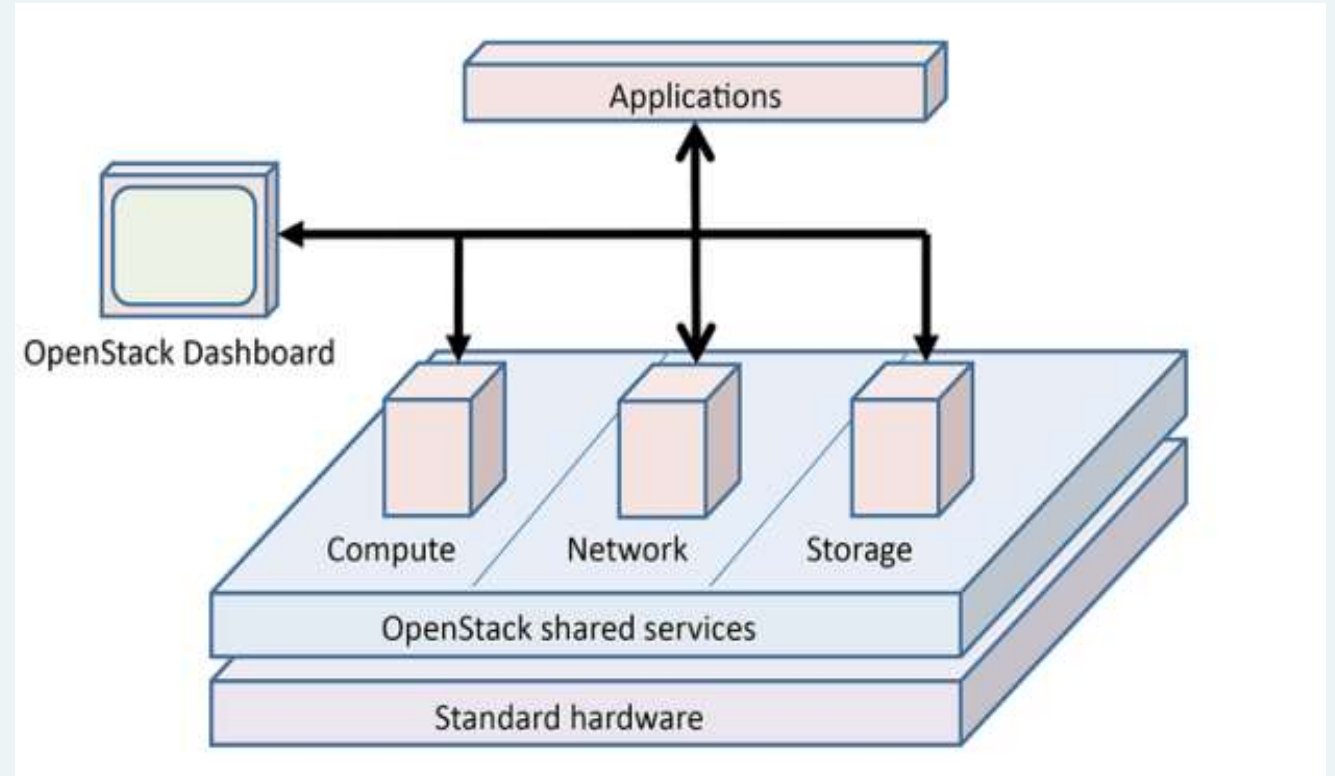
- The northbound interface between the application plane and controller passes information pertaining to the applications' needs so that the controller can open up the best possible software network with the appropriate QoS, adequate security and the essential management for the operations to be able to be carried with no problems.
- The basic protocol is based on the REST (Representative State Transfer) API. This interface must enable us to send the information necessary for the configuration, the operations and the measurement report. This RESTful protocol is integrated into the interface of numerous Cloud management systems such as:
 - OpenStack, (REST API is integrated), those of the service providers, and the API Virtual Private Cloud (VPC).
- The representations facilitated by the REST protocol enable us to pass the information over the northbound interface with the following properties:
 - each resource is identified individually, the resources can be manipulated by representations,
 - the messages are self-descriptive: they explain their nature by themselves, each access to the subsequent states of the application is described in the present message.

Application Layer

- The application layer is essentially formed of Clouds that contain the application virtual machines. These may be business machines or network element management machines – responsible, for instance, for the managing of handovers or determination of the best access, at any given time, for a multi-technology terminal.
- This layer essentially contains the operating systems for the Cloud. The best-known such system is, undoubtedly, **OpenStack**. It is a Cloud management system that controls large sets of resource offering processing power, storage and network resources.
- OpenStack has already had more than twenty releases in only nine years (start of 2011 to start of 2019).
- **Stein** was released at the start of 2019. OpenStack is open source software with an Apache license.
- It is modular, and contains numerous modules, such as **Nova** for computation, **Swift** for storage, **Glance** for the imaging service, **Dashboard** for the settings and control panel.

OpenStack System

- OpenStack provides flexible network models to take care of the applications.
- **OpenStack Neutron** (Network module) manages the IP addresses and allows for static addresses or uses DHCP. Users can create their own network in which SDN technology is used. OpenStack has numerous extensions, such as IDS (Intrusion Detection Systems), load balancing, the option to deploy firewalls and VPNs.

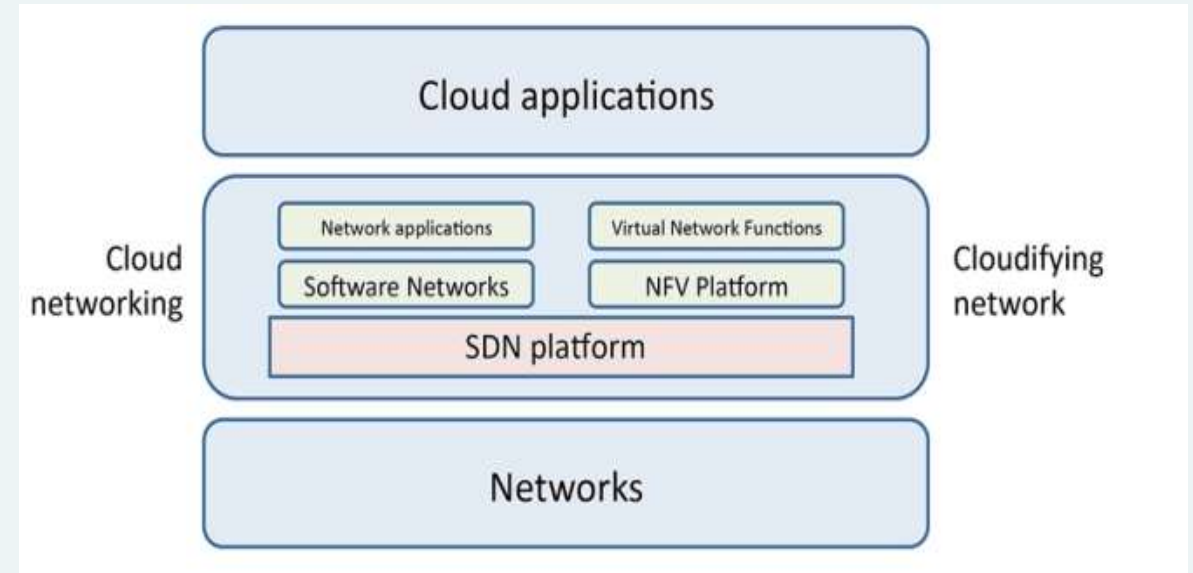


Summary of SDN Architectures

- The top and bottom parts represent the Cloud and the physical/logical networks. Between these two points, management and control of the network and of the applications needs to take place. In terms of the business application, we find sets of software modules – mostly open source – to deploy cloud-computing infrastructures, and more specifically IaaS (Infrastructure as a Service).
- On the contrary, we find the applications to establish a virtualized network structure, with the commands necessary to handle the business applications.

SDN Solutions

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Exercise

- How does an SDN differ from a traditional network in terms of configuration and management?
- In what ways can SDN simplify network operations compared to traditional networking approaches?
- What are some security vulnerabilities or risks introduced by SDN?
- Discuss methods for protecting the SDN controller against threats.



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